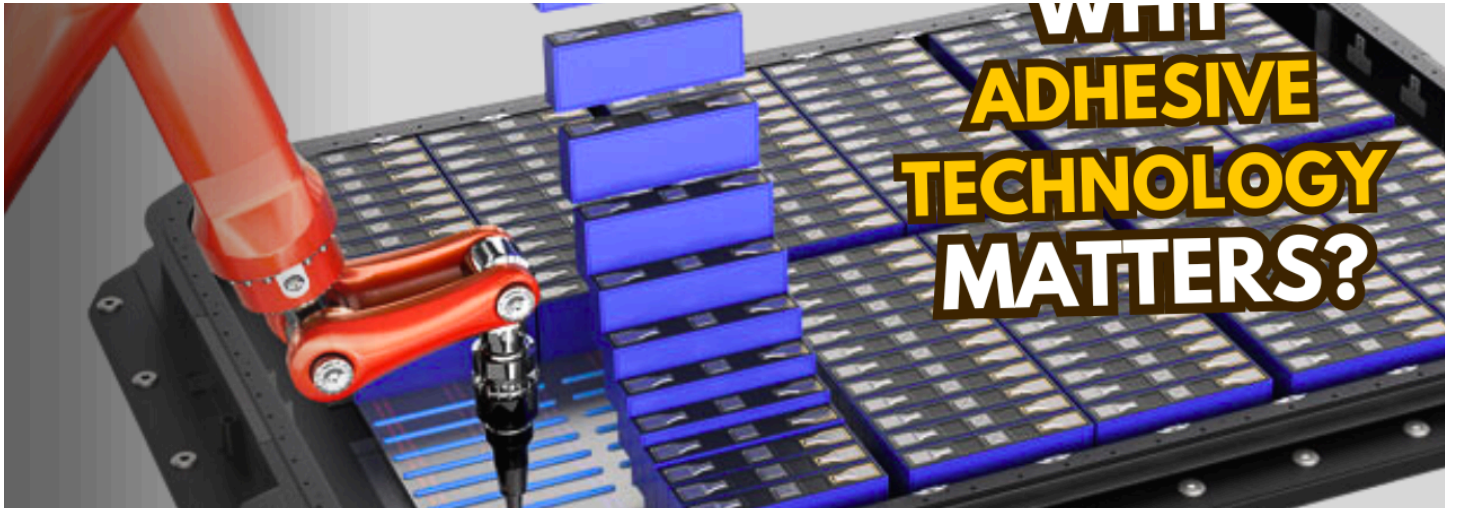




Why Adhesives Technology for EV Batteries Matters More Than Ever



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Introduction

The electric vehicle (EV) industry is rapidly expanding, with a growing emphasis on battery safety, performance, and durability. Adhesives are playing an increasingly critical role in securing and protecting EV batteries, contributing to improved thermal management, enhanced durability, and overall vehicle performance. As the demand for EVs continues to rise, advancements in adhesive technology are becoming ever more important.

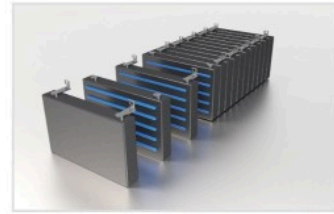
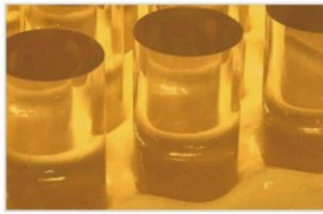
The Vital Role of Adhesives in EV Batteries

Securing and Protecting: Adhesives are essential for holding EV battery components together, distributing stress, and preventing vibration. They secure the battery in place and protect it from external factors that could cause damage.

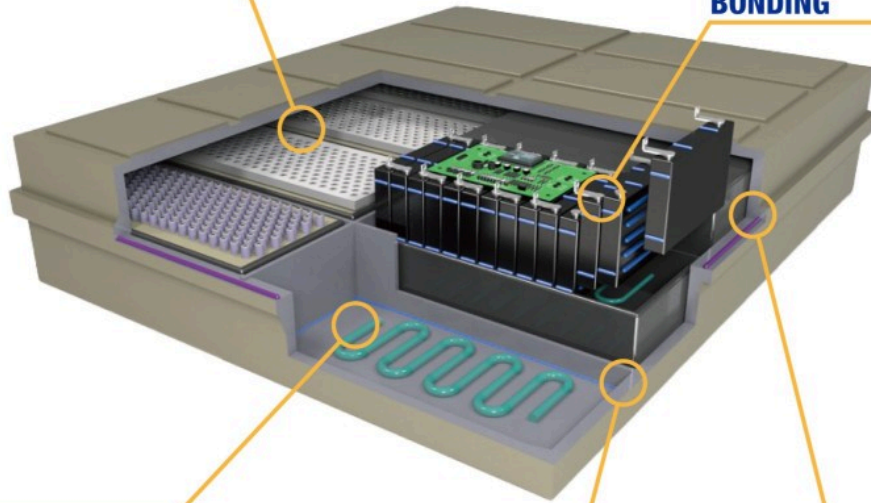
Improved Thermal Management: Adhesives help dissipate heat effectively, preventing battery overheating, which can lead to degradation and safety issues. Thermally conductive adhesives and interface materials support heat dissipation, ensuring consistent performance and extending battery lifespan.



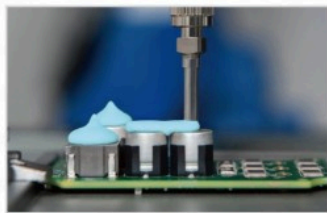
**CELL TO CARRIER BONDING
AND MODULE ENCAPSULATION**



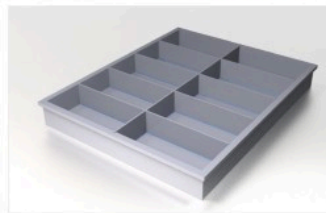
**CELL TO CELL
& CELL TO FRAME
BONDING**



**THERMAL MANAGEMENT
- GAP FILLING**



STRUCTURAL BONDING



PACK SEALING

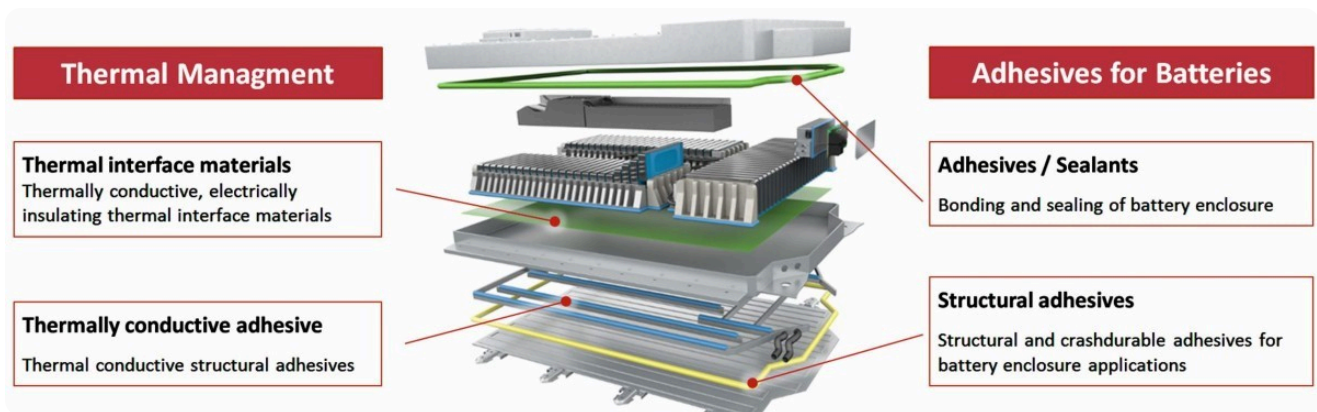


Enhanced Durability: EV batteries are subject to vibrations, impacts, and constant movement. Adhesives provide the flexibility and toughness needed to withstand these stresses, preventing cracking and other forms of damage.

Design Flexibility: Adhesives enable greater design flexibility by bonding various materials, including composites, functional films, and metals. This allows for innovative battery designs that were previously unattainable.

Types of Adhesives Used in EV Batteries

Structural adhesives: Structural adhesives are high-strength adhesives that are used to bond battery components together, such as cells, modules, and packs. They provide a strong and durable bond that can withstand the stresses and vibrations associated with EV operation. Common types of structural adhesives used in EV batteries include Epoxy adhesives, Polyurethane adhesives and Acrylic adhesives.



Cooling plate assemblies: Cooling systems are vital for maintaining the optimal temperature of battery cells in an EV. Adhesives join cooling plate assemblies, often combining hybrid materials like plastic and metal. These adhesives provide structural support and seal the water glycol fluid used for cooling. This innovative approach results in lightweight and efficient cooling plates that contribute to the battery's overall performance.

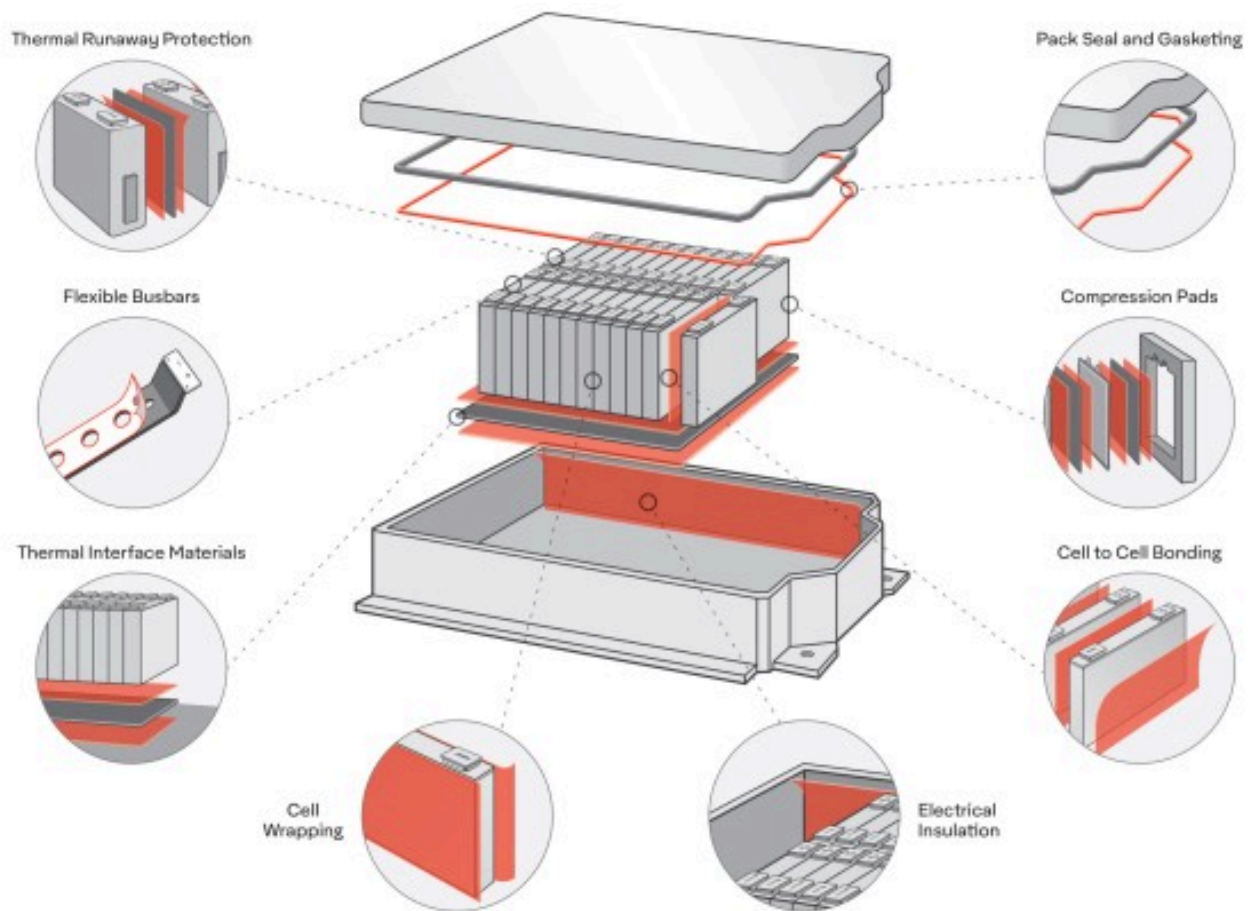
Thermal interface materials: In the upper part of the battery pack, thermal interface materials, often gap fillers, play a critical role in heat transfer and thermal management. According to Billotto, these adhesive materials act as interfaces between the battery cells and the cooling plates, ensuring heat is efficiently dissipated during charging and discharging. These adhesives enhance battery longevity by helping keep the batteries within the optimal temperature range (typically 35-60°C).

Thermal management and encapsulation

To mitigate the risk of thermal runaway in individual cells, adhesives are used to bond cells together and are encapsulated with polyurethane material. This encapsulation prevents the spread of thermal events from one cell to the next, adding an essential layer of safety to EV battery packs. In some designs, cylindrical cells are bonded to cooling ribbons, which are thermally conductive and structurally adhesive.

Cell-to-pack designs and cell-to-cell bonding

The next generation of EV batteries is witnessing the emergence of cell-to-pack designs. These designs integrate battery cells into the pack using thermal structural adhesives. This multifunctional approach enhances the battery's performance and contributes to safety.

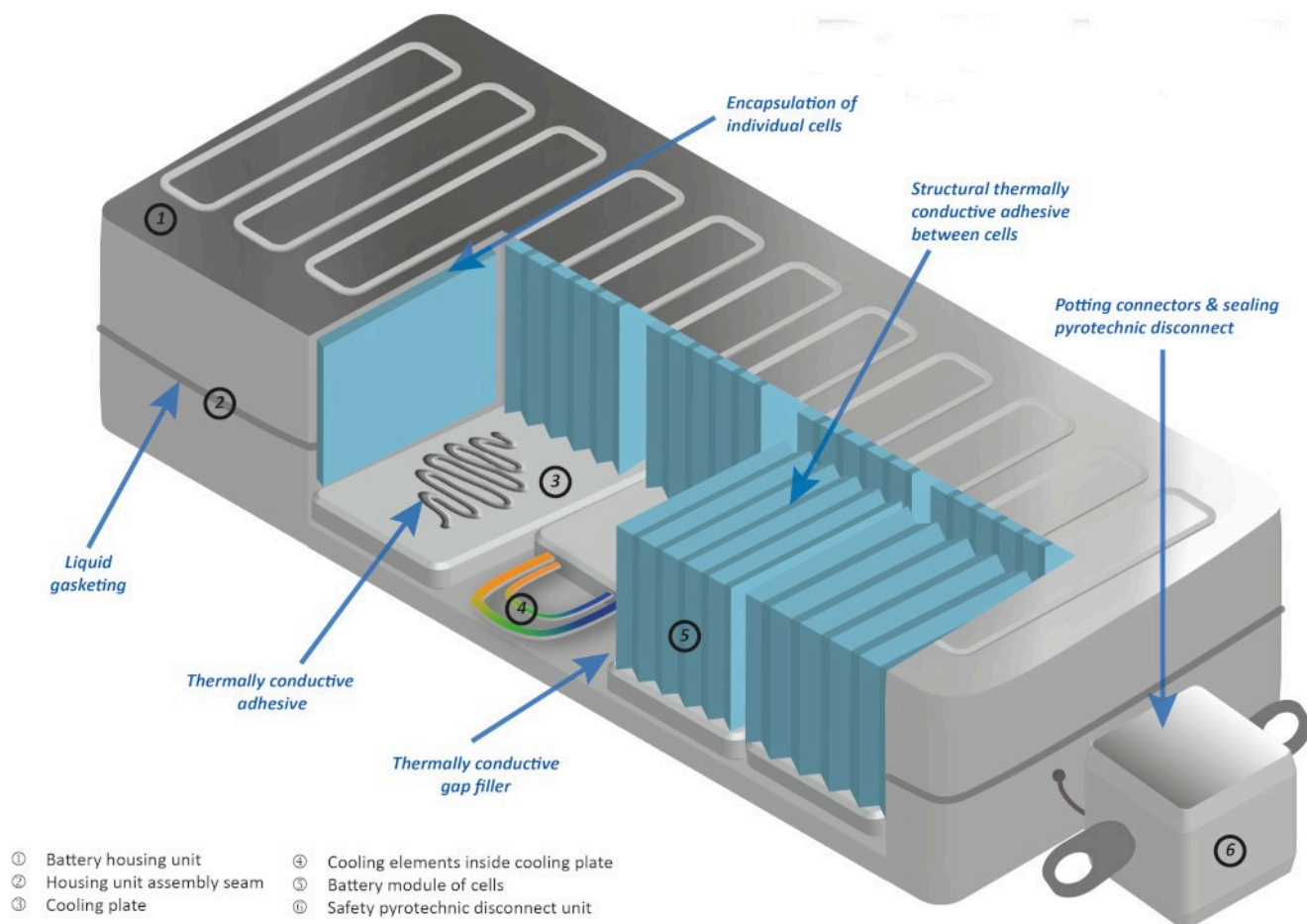


Qualities to look for in adhesives and sealants for battery assembly

In addition to performance, EV battery designers know that adhesives and sealants must work well in high-volume production. An application must be efficient and allow for a repeatable process.

Specific manufacturing qualities to look for in adhesives, sealants, and thermal interface materials include:

- Non-abrasive formulation to avoid equipment wear
- Compatibility with fast dispensing
- Low press-in force for fast joining
- Low pullout force for repairability
- Good contact/no gaps between battery cells and cooling plate for retention of thermal conductivity
- No physical changes over aging
- No sagging in vertical aging tests
- Long working time of 30 minutes or more



Key Benefits of Adhesives in EV Batteries

- **Bonding battery components:** Adhesives can be used to bond battery cells, modules, and packs together, creating a strong and durable structure.
- **Improving thermal management:** Adhesives can be used to create a thermal interface between battery components, helping to dissipate heat and keep the battery cool.
- **Enhancing safety:** Adhesives can be used to create a barrier between battery cells, preventing the spread of thermal runaway in the event of a cell failure.
- **Reducing weight:** Adhesives can be used to replace traditional fasteners, such as screws and bolts, which can help to reduce the weight of the battery.

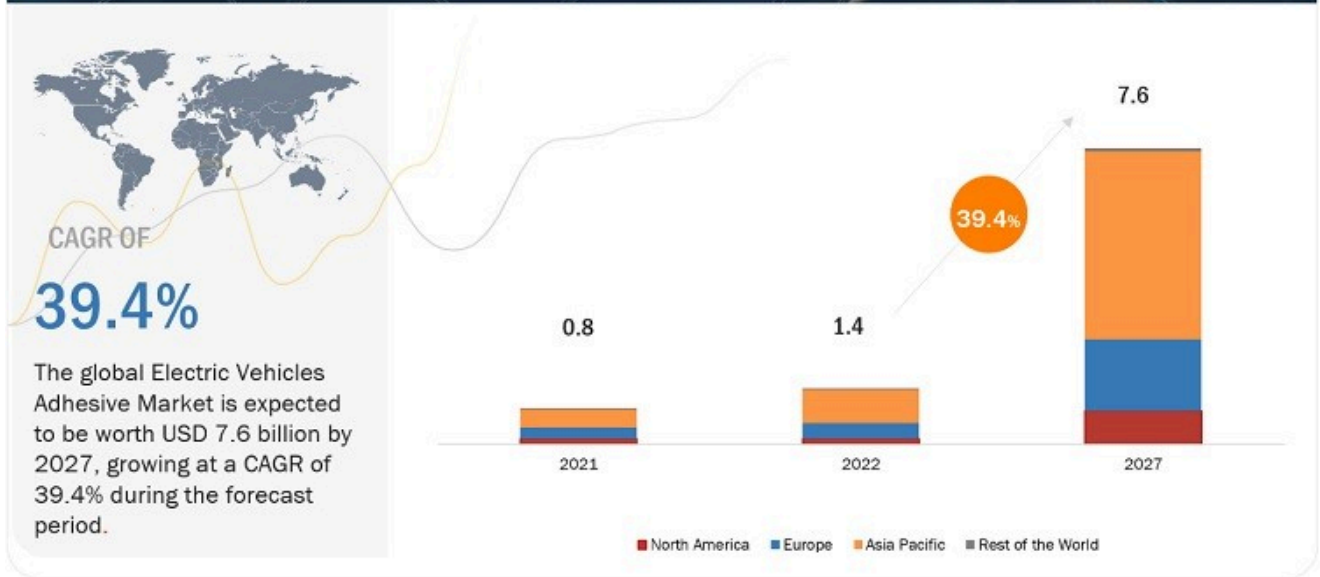
Sustainability and Cost Savings

Lightweight Design: Adhesives contribute to sustainability by enabling lightweight vehicle designs and reducing panel thickness in body structures, resulting in significant CO2 savings.

Streamlined Manufacturing: Adhesives streamline manufacturing processes by offering fast, low-cost joining and curing, eliminating the need for surface pretreatment and improving process speed.

ELECTRIC VEHICLE ADHESIVES MARKET

GLOBAL FORECAST TO 2027 (USD BN)



Conclusion

As EV adoption grows, adhesives are becoming **mission-critical** for battery safety, efficiency, and durability. With innovations in **nanotechnology**, **smart adhesives**, and **sustainability**, the future looks promising for adhesive solutions in EV batteries.

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